

Nested Transfer and Scale-Free Spectra

The superposition origin of $1/f$, its position within a one-parameter family, and five domains read as one relational configuration

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Lab notes — record of independent work. No claim of ownership or discovery.

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Abstract

This paper fixes an exact finite-band superposition formula for exponentially correlated contributions distributed across relaxation times. When the effective spectral-weight density satisfies $g(\tau) = A\tau^{-\gamma}$, $0 < \gamma < 2$, its interior-band spectrum scales as $f^{\gamma-2}$; equal effective weight per logarithmic interval, $g(\tau) \propto 1/\tau$, gives $1/f$. The mathematics is standard. The offered contribution is a frame-marked reading of turbulence, accretion-disc variability, atmospheric-Cherenkov lateral distributions, trapped-charge noise, and threshold-release systems as registrations of one nested-transfer configuration without reducing them to one microscopic cause or privileging one readout as fundamental. Local and global descriptions, object and mirror, closure and reopenability become identifiable through the registry that couples them. The paper separates exact within-registry derivations from cross-registry correspondence, states limitations and counterexamples, and ends with finite falsification surfaces and a residue ledger rather than an assertion of completed ontological closure.

Tier discipline

Tier I contains standard, checkable mathematics with derivation or citation. **Tier II** reads structure after the Tier-I objects and their registries have been fixed. **Tier III** declares a direction of inquiry. A Tier-II correspondence does not silently inherit the proof status of the Tier-I results it relates. Falsification surfaces appear in §10.

Contents

1	Origin, standing, and method	3
1.1	Infinite registry and recursive signature	4
1.2	A geometric scaffold: polygonal closure, Clifford torus, and Hopf transfer	4
1.2.1	Finite planar closure and incremental refinement	4
1.2.2	The Clifford-torus carrier	4
1.2.3	Hopf-fibered registry	5
2	Primitives and the relational configuration	6
3	The superposition theorem	6
3.1	Exact finite-band statement	7
3.2	The logarithmic point and finite corners	7
3.3	Relational reading	8
4	Projection, homogeneity, and residue	8
5	Worked registrations	9
5.1	Turbulent cascade	9
5.2	Accretion-disc variability	9

5.3	Atmospheric-Cherenkov lateral distributions	10
5.4	Trapped-charge noise	11
5.5	Threshold-release systems	11
6	Contrasts and exclusions	11
6.1	Johnson–Nyquist noise	12
6.2	Independent sampling	12
6.3	Brownian motion	12
6.4	BTW temporal activity	12
7	What is claimed	13
8	Exponent ledger	13
9	Observer, mirror, and self-definition	14
10	Falsification surfaces	14
11	Driven consequences: the registry ladder as a $1/f$ readout	15
11.1	A geometric ladder of frames	15
11.2	Why stopping produces both meaning and finite power	16
11.3	Ordered comparison without an exterior simultaneous view	17
11.4	Perturbation as the self/other seam	17
11.5	Structural consequence	17
12	Residue ledger and executable continuations	18
12.1	Distributed calibration, not shared experience	18
12.2	Correlated contributions: an explicit exponent rule	18
12.3	Three operational readout classes	19
12.4	A one-step Cherenkov tail construction	19
12.5	Resolution is indexed	20
12.6	Discrete and continuous scale registration	20
12.7	A partial transition-map result	20
12.8	Recursive frame edges remain plural	20
	References	21

1. Origin, standing, and method

The originating observation concerned the shape of quantities registered across nested levels toward a boundary and the form acquired by a readout that resolves those relations. Subsequent checking located established mathematics: the relaxation-time superposition associated with Bernamont and later noise theory; the generalized central-limit framework of Gnedenko and Kolmogorov; Kolmogorov’s inertial-range treatment; Lyubarskii’s disc model; and the McWhorter trapped-charge model. The mathematics is not presented as new.

The author has not located a prior treatment reading the five domains assembled here as one configuration. This remains a report of search, not a proof of absence. The paper offers the reading, states exactly what its mathematics establishes, and exposes the points at which the reading can fail.

Intellectual lineage and scope of TUFT

The TUFT name and the Hopf-fibered structural lineage used in the author’s later Λ -TUFT-NSAF work require explicit attribution to **Jenny Lorraine Nielsen**. Nielsen’s *The Topological Unified Field Theory on the Complex Hopf Fibration* is her independently conceived, authored, and owned academic work. Her TUFT and the subsequent collaborative exchange around it were foundational to the author’s later constructions and should be credited wherever that structural lineage is used.

This paper makes a narrower claim. It uses the complex Hopf fibration as standard mathematics and as a registry scaffold for the relational reading developed here. It does not reproduce or rely upon Nielsen’s wider physical derivations, does not adjudicate them, and neither refutes nor takes exception to them. Citation records intellectual lineage and preserves authorship; it does not assign Nielsen responsibility for this paper’s claims or imply her endorsement of them.

Registry discipline

Every object, level, scale, boundary, transfer, and readout used here is fixed within a declared descriptive registry. Boolean distinctions and exact mathematical relations remain valid where their predicates, domains, and admissible transformations have been fixed. Their validity does not establish that the resulting partition exhausts the configuration, that one term possesses observer-independent fundamentality, or that an arrow oriented for calculation retains that orientation outside the registry.

When a requested description needs degrees of freedom absent from the stated model, the paper records a frame-edge event. It does not convert representational insufficiency into evidence for a specially witnessed transcendent object. A binary gate may close a predicate without closing the configuration from which the predicate was selected.

The paper therefore distinguishes mathematical direction from ontological priority. An equation written as $g \mapsto S$ may support forward calculation; an inverse problem may read $S \mapsto g$. Neither orientation alone establishes how reality is ordered independently of the registry. More generally, every modeled direction — forward, inverse, causal, reconstructive, or predictive — assigns primitive and derived roles inside a description.

1.1 Infinite registry and recursive signature

The registry is not first finite and then indefinitely iterated, nor first a completed infinity from which finite iterations derive. Infinite potential and iterable recurrence become identifiable together. Without reopenable iteration, infinite potential has no registry-internal articulation; without infinite potential, iteration terminates as a closed procedure. Neither relation receives primitive status.

A useful formal image is a recursive relation

$$\mathcal{R} \simeq \mathfrak{F}(\mathcal{R}),$$

not as a reduction of $\mathcal{R}$ to a more fundamental object, but as a mark of self-including expandability: a finite closure may re-enter as content, boundary, or transformation in another articulation. The signature is the repeated failure of any station, including one that names the registry, to establish singular fundamentality over the relation that made the naming possible.

1.2 A geometric scaffold: polygonal closure, Clifford torus, and Hopf transfer

The preceding registry is not intended as an uninstantiated infinity awaiting contents. In the author's standard Λ -TUFT-NSAF construction, it receives an explicit geometric scaffold. The claim made here is sufficiency, not uniqueness or historical priority: the construction demonstrably supports finite closure, indefinite refinement, phase-bearing fibers, dimensional extension, projection, and recursive return. A prior or more economical scaffold would not contradict this use.

Finite planar closure and incremental refinement

In a Euclidean plane, three non-collinear vertices provide the least polygonal closure: a triangle. For a unit circle, the regular inscribed n -gon, where n is the number of sides, has vertices

$$v_k = \exp\left(\frac{2\pi i k}{n}\right), \quad k = 0, \dots, n-1,$$

and perimeter

$$P_n = 2n \sin \frac{\pi}{n} \longrightarrow 2\pi \quad (n \rightarrow \infty).$$

Its vertex and chord relations form a finite grid at every n , while the family admits unbounded angular refinement. No finite polygon becomes the circle; each gives an exact finite closure under its own incidence relations. The passage $n \mapsto n+1$, or to any chosen refinement sequence, changes the resolving registry rather than revealing a final privileged polygon.

This supplies an early step *within the stated construction*: distinction closes locally, the closure can be reinscribed, and the residue between polygon and circle remains available for another articulation. It does not assert that the triangle or polygonal sequence constitutes an observer-independent first object.

The Clifford-torus carrier

Write

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}.$$

The Clifford torus

$$T_{\text{Cl}} = \left\{ (z_1, z_2) \in S^3 : |z_1| = |z_2| = \frac{1}{\sqrt{2}} \right\} \cong S^1 \times S^1$$

is an exact two-phase carrier inside S^3 . Under the diagonal circle action

$$(z_1, z_2) \mapsto (e^{i\theta} z_1, e^{i\theta} z_2),$$

the torus is foliated by Hopf fibers; the relative phase $\arg z_1 - \arg z_2$ distinguishes the fiber orbits within the torus. Under the standard Hopf map $S^3 \rightarrow S^2$, the Clifford torus projects to the equator of the base sphere. These are Tier-I geometric facts.

The author's earlier "Klein-like" use of the Clifford torus is retained here only as a constructional reading: a lower-dimensional closure is carried through identification, twist, and projection into a higher-dimensional relation without claiming that the higher order causally derives from the lower or that the lower is merely a shadow of a completed higher object. The torus functions as an intermediate stair, not as a declared universal origin.

Hopf-fibered registry

The circle Hopf bundles are

$$S^1 \hookrightarrow S^{2d+1} \xrightarrow{\pi_H} \mathbb{CP}^d, \quad d \geq 1, \quad (1)$$

with the basic case

$$S^1 \hookrightarrow S^3 \xrightarrow{\pi_H} S^2 \cong \mathbb{CP}^1.$$

A point of $\mathbb{CP}^d$ is a complex line in $\mathbb{C}^{d+1}$; its fiber consists of the common phases $e^{i\theta} z$ of a unit representative $z \in S^{2d+1}$. Base, fiber, and total space therefore give three non-interchangeable but coupled registrations: projective relation, phase, and their joined carrier.

In the Λ -TUFT-NSAF reading, equation (1) functions as an Infinite Registry scaffold because it combines:

- finite projective or polygonal statements at a declared resolution;
- a continuous S^1 phase relation not exhausted by any one finite grid;
- extension through $d = 1, 2, \dots$ without declaring a completed terminal dimension;
- lift, twist, and reattachment between base descriptions and fiber phases;
- linked fibers and recursive re-description of earlier closures.

The projected inscribed n -gon grids and the dimensional index d must not be conflated: n is the number of polygon sides and controls finite angular resolution within a chosen projection, while d controls the complex-projective carrier. Their joint variation supplies incremental descriptive freedom in two distinct registry directions.

Gauge theory supplies a standard mathematical language for this transfer. A gauge field is formulated through a principal bundle, local sections, a connection, curvature, and holonomy; local representatives related by gauge transformation describe one bundle relation without becoming identical coordinate expressions. The Hopf bundle is a canonical $U(1)$ prototype. The structural claim here is therefore not that all gauge theories reduce to polygons or to this one bundle. It is that the same topological operations—local section, overlap, phase transformation, lift, and projected invariant—demonstrate how meaningful statements can transfer among registry patches without installing a single patch as

transcendent authority.

The construction resembles the IIRPrOOF engine in method: begin with declared primitives, close the least relation presently required, register the residue, and permit that closure to become a primitive in a further lift. A single planar closure is to the recursively fibered construction as one stair is to an escalator: distinguishable as a local step, yet participating in a traversal whose iterability and open extent become identifiable together. This is a Tier-II correspondence and a statement of the author's construction path, not a claim that the path has already been proven unique or exhaustive.

2. Primitives and the relational configuration

P1 — Accounted quantity. A quantity Q is registered as held at, passed among, entering, or leaving declared levels. Conservation may be imposed in a case; the general configuration requires only that unmodeled creation not be silently inserted into the stated balance. This is an accounting convention, not a claim about an absolute origin of Q .

P2 — Nested levels. An ordered family $\{\ell_s\}$ is indexed by a scale coordinate s fixed by the registry. Each level has a characteristic time $\tau(s)$, and its registered state is conditioned by the traversal through other levels. “Higher,” “lower,” “local,” and “global” refer to this ordering, not to ontological rank.

P3 — Discharge boundary. A level at which Q leaves the modeled hierarchy by dissipation, accretion, radiation, release, or another declared operation. What functions as discharge in one registry may function as injection or persistence in another. The boundary makes the stated traversal non-circulatory without claiming an observer-free terminal end.

P4 — Scale-free range. Relative to the selected coordinate s , transformation class $s \mapsto \lambda s$, and readout-relevant equivalence, no level in $[s_{\min}, s_{\max}]$ is distinguished by the transfer relation. This is the substantive hypothesis. It asserts registry-relative invariance, not the absence of distinguished structure under every description.

P5 — Readout. A functional through which relations become observable: temporal or wavenumber spectrum, spatial density, structure function, event-size distribution, or another declared projection. A readout does not merely reveal a completed object from outside; it helps fix which relation becomes the observable under consideration.

Definition 2.1 (Nested-transfer configuration). A declared registry satisfying P1–P4, together with at least one readout P5 when an observable claim is made.

The local side does not simply cause the global side. Nor does the global side exist independently and then cause its local manifestations. Each becomes identifiable through the registry that couples them. A compact representation is

$$\mathcal{C} = (\text{levels, transfer, scale measure, boundary, readout}).$$

Its projections $\Pi_{\text{local}}(\mathcal{C})$ and $\Pi_{\text{global}}(\mathcal{C})$ remain non-identical but non-independent within the declared relation.

3. The superposition theorem

3.1 Exact finite-band statement

For clarity, $g(\tau)$ below is an *effective spectral-weight density*. It incorporates both the number of contributions near τ and their readout variance. If a number density $n(\tau)$ and variance $\sigma^2(\tau)$ are separately specified, then $g(\tau) = n(\tau)\sigma^2(\tau)$.

Theorem 3.1 (Finite-band relaxation superposition; Tier I). *Suppose zero-mean, mutually uncorrelated stationary contributions have autocovariance shape $e^{-|t|/\tau}$ and effective spectral-weight density*

$$g(\tau) = A\tau^{-\gamma}, \quad \tau \in [\tau_{\min}, \tau_{\max}], \quad 0 < \gamma < 2.$$

Under the two-sided Fourier convention with frequency f in cycles per unit time, their summed power spectral density is exactly

$$S(f) = 2A(2\pi|f|)^{\gamma-2} \int_{2\pi|f|\tau_{\min}}^{2\pi|f|\tau_{\max}} \frac{u^{1-\gamma}}{1+u^2} du. \quad (2)$$

In the interior band

$$2\pi|f|\tau_{\min} \ll 1 \ll 2\pi|f|\tau_{\max},$$

the asymptotic form is

$$S(f) \sim \frac{A\pi}{\sin(\pi\gamma/2)} (2\pi|f|)^{\gamma-2}. \quad (3)$$

For $\gamma = 1$, $S(f) \sim A/(2|f|)$.

Proof. A unit-variance exponential autocovariance has Lorentzian spectrum

$$K(f, \tau) = \int_{-\infty}^{\infty} e^{-|t|/\tau} e^{-i2\pi ft} dt = \frac{2\tau}{1 + (2\pi f\tau)^2}.$$

Uncorrelated contributions add in power, hence

$$S(f) = \int_{\tau_{\min}}^{\tau_{\max}} \frac{2\tau}{1 + (2\pi f\tau)^2} A\tau^{-\gamma} d\tau.$$

Set $u = 2\pi|f|\tau$. This gives (2). In the interior band, the endpoints may be extended asymptotically to zero and infinity. With $s = 2 - \gamma$,

$$\int_0^{\infty} \frac{u^{s-1}}{1+u^2} du = \frac{\pi}{2 \sin(\pi s/2)}, \quad 0 < s < 2.$$

Since $\sin[\pi(2 - \gamma)/2] = \sin(\pi\gamma/2)$, equation (3) follows. $\square$

3.2 The logarithmic point and finite corners

For $\gamma = 1$, the exact finite-band expression is

$$S(f) = \frac{A}{\pi|f|} [\arctan(2\pi|f|\tau_{\max}) - \arctan(2\pi|f|\tau_{\min})]. \quad (4)$$

At low frequency, $S(f) \rightarrow 2A(\tau_{\max} - \tau_{\min})$. At high frequency,

$$S(f) \sim \frac{A}{2\pi^2 f^2} \left(\frac{1}{\tau_{\min}} - \frac{1}{\tau_{\max}} \right).$$

The characteristic crossovers lie near $(2\pi\tau_{\max})^{-1}$ and $(2\pi\tau_{\min})^{-1}$; they are smooth crossovers, not mathematically sharp corners.

At $\gamma = 0$ or 2 , the Mellin integral ceases to converge at an endpoint. Finite cutoffs then produce logarithmic corrections to nominal f^{-2} or white behavior. These boundary cases do not belong to Theorem 3.1's open interval.

Equal effective weight per logarithmic interval means

$$g(\tau) d\tau \propto d \log \tau = \frac{d\tau}{\tau}.$$

It is exact relative to the selected measure. It does not establish logarithmic time as an observer-independent fundamental coordinate.

3.3 Relational reading

Define

$$\mathfrak{R}_K = \left\{ (g, S) : S(f) = \int K(f, \tau) g(\tau) d\tau \right\}.$$

The theorem fixes a relation between two projections. Forward evaluation may orient it as $g \mapsto S$; an inverse problem may orient it as $S \mapsto g$. Neither orientation converts calculation into ontological priority. Superposition here means joint participation in a readout, not the metaphysical claim that many lower causes create one derivative whole.

4. Projection, homogeneity, and residue

Proposition 4.1 (Conditional projection principle; Tier I/II boundary). *Let a registered readout inherit continuous homogeneity over a declared range and introduce no additional scale there. If*

$$R_{\mathcal{C}}(\lambda x) = \lambda^{-\kappa} R_{\mathcal{C}}(x)$$

for every admissible $\lambda > 0$, then, under ordinary measurability or continuity assumptions,

$$R_{\mathcal{C}}(x) = Cx^{-\kappa}.$$

The exponent belongs to the coupled configuration/readout relation, not to either projection independently.

Scale-freedom of an underlying relation does not force every conceivable readout to inherit exact homogeneity. A readout may introduce filtering, saturation, convolution, another characteristic scale, or discrete-scale modulation. The proposition therefore states its inheritance condition explicitly.

For an expression or readout E , the same gate that releases a determinate result leaves a residue $\Lambda(E)$

of relations not carried by that result. Thus

$$\text{meaningful closure} \quad \not\Rightarrow \quad \text{expressive residue}.$$

A whole expression includes its frame and boundary; wholeness does not mean exhaustive description.

5. Worked registrations

Each case contains a Tier-I result internal to its own registry and a Tier-II comparison. The comparison claims preservation of a stated organization, not identity of microscopic mechanism.

5.1 Turbulent cascade

Registry. Q : kinetic energy per unit mass; levels: eddy scale ℓ or wavenumber k ; transfer: mean flux ε ; discharge: viscous dissipation near $\eta = (\nu^3/\varepsilon)^{1/4}$; range: $\eta \ll \ell \ll L$.

For stationary, homogeneous, isotropic turbulence in the inertial-range/high-Reynolds-number limit, the four-fifths law gives the exact third-order longitudinal relation

$$S_3(r) = \langle (\delta u_{\parallel})^3 \rangle = -\frac{4}{5}\varepsilon r.$$

This fixes the mean flux in that registry. Under the additional K41 similarity assumption that $E(k)$ depends only on ε and k , dimensional analysis yields

$$E(k) = C_K \varepsilon^{2/3} k^{-5/3}.$$

The dimensionless constant C_K is not fixed by dimensional analysis, and intermittency limits any claim that every structure-function exponent equals its K41 value exactly. The second-order K41 prediction is

$$S_2(r) \propto (\varepsilon r)^{2/3}.$$

Under Taylor's frozen-flow hypothesis, $k = 2\pi f/U$ and

$$S_f(f) = E_k(k) \left| \frac{dk}{df} \right| \propto f^{-5/3}.$$

Tier-II reading. A constant-flux registration and different readouts generate different exponents without contradiction. This case does not reduce to Theorem 3.1; it mirrors the principle that exponent and readout are coupled.

5.2 Accretion-disc variability

Registry. Q : mass; levels: annuli at radius r ; traversal: inward accretion and propagated fluctuations; discharge: the modeled inner flow boundary; readout: luminosity or accretion-rate spectrum.

For a thin-disc scaling,

$$f_{\text{visc}}(r) \sim \alpha_{\text{ss}} \left(\frac{H}{r} \right)^2 \Omega_K(r), \quad \Omega_K \propto r^{-3/2}.$$

If α_{ss} and H/r are approximately constant, then

$$\tau_{\text{visc}}(r) \propto r^{3/2}, \quad d \log \tau_{\text{visc}} = \frac{3}{2} d \log r.$$

Equal effective fluctuation power per logarithmic radial interval therefore corresponds to equal effective weight per logarithmic viscous-time interval. Lyubarskii independently derives flicker noise from a linearized disc-diffusion description when viscosity fluctuations at separated radial scales are uncorrelated, occur on local viscous times, and have suitable radial amplitude.

Tier-II reading. The diffusion calculation and the independent-relaxator theorem are distinct descriptions. Their preserved relation is logarithmic scale organization. Neither must causally reduce to the other for the comparison to hold. The observable band is associated with outer and inner effective fluctuation-generating radii; identifying the latter with an ISCO requires an additional black-hole disc model and is not universal.

5.3 Atmospheric-Cherenkov lateral distributions

Registry. Q : shower energy registered through particle and photon populations; levels: cascade generations; boundaries: first interaction, shower development, critical-energy termination, and observation plane; readout: radial Cherenkov-photon density.

The generalized central-limit theorem states that if iid variables satisfy

$$\Pr(|X| > x) \sim x^{-\alpha} L(x), \quad 0 < \alpha < 2,$$

with L slowly varying, normalized sums converge to an α -stable law. For $\alpha \neq 1$, one common parameterization has characteristic function

$$\varphi(t) = \exp \left[-|ct|^\alpha \left(1 - i\beta \operatorname{sgn}(t) \tan \frac{\pi\alpha}{2} \right) + i\mu t \right].$$

Its one-dimensional density has tails proportional to $|x|^{-(1+\alpha)}$, with side-dependent coefficients.

Volkov, Lagutin, Reviakin, and Bizhanov fit CORSIKA Cherenkov lateral distributions using one-dimensional fractional-stable densities and report break scales and energy-dependent fitted α, β parameters. Their work establishes an empirical fitting representation; it does not derive the fitted stable index from microscopic Rutherford scattering. Atomic screening and physical cutoffs also prevent a bare small-angle divergence from serving, by itself, as proof of infinite variance. Whether a one-dimensional density-tail exponent transfers directly to a reported radial LDF depends on the paper's radial-density convention.

For vertical proton showers, their Table 1 reports

$$(\alpha, \beta, R_{\text{Ch}}) = (1.21, 0.75, 81.1 \text{ m}) \quad \text{at} \quad 10^4 \text{ GeV},$$

and $(1.33, 0.11, 32.4 \text{ m})$ at 10^9 GeV , with intermediate values tabulated rather than inferred here. For the simulated electron-radius/break relation they report

$$R_e = (0.77 \pm 0.03) R_{\text{Ch}} + (83.11 \pm 2.34) \text{ m}.$$

They also display an SD fit to TAIGA–HiSCORE data. These numbers are reported fit results, not

first-principles predictions of this paper.

Tier-II reading. The break and power-law asymptotes provide a spatial mirror of bounded scale organization. Convergence toward a non-Gaussian stable attractor with increasing shower development remains a testable interpretation, not a Tier-I derivation supplied here.

5.4 Trapped-charge noise

Registry. Q : charge; levels: trap energies E or depths x ; operations: capture and emission; discharge: coupling to a conduction channel; readout: current or voltage spectrum.

Thermal activation and tunneling give, respectively,

$$\tau(E) = \tau_0 e^{E/(k_B T)}, \quad \tau(x) = \tau_0 e^{x/\lambda}.$$

Hence

$$dE = k_B T \frac{d\tau}{\tau}, \quad dx = \lambda \frac{d\tau}{\tau}.$$

A uniform density in E or x , together with approximately uniform effective coupling to the readout, gives effective spectral weight $g(\tau) \propto 1/\tau$. Theorem 3.1 then yields an interior $1/f$ band with cutoff-controlled crossovers. This is the clean control case: the logarithmic weighting follows from a stated coordinate relation, while departures in trap density or coupling predict departures from $1/f$.

5.5 Threshold-release systems

Registry. Q : grains, elastic stress, or magnetic free energy; levels: regions or events over a size range; gate: a release threshold; discharge: avalanche, rupture, or flare; readouts: event size, duration, or temporal activity.

For earthquakes,

$$\log_{10} N(> M) = a - bM, \quad E \propto 10^{1.5M},$$

so

$$N(> E) \propto E^{-b/1.5},$$

which is approximately $E^{-2/3}$ when $b \approx 1$.

Bak, Tang, and Wiesenfeld proposed self-organized criticality as an account of $1/f$ noise. Jensen, Christensen, and Fogedby found a temporal spectrum near f^{-2} for BTW sand flow. That exponent is not derived here from Theorem 3.1: an exact -2 would correspond to the excluded endpoint $\gamma = 0$, generally with logarithmic corrections, and avalanche pulses need not be independent exponential relaxators.

Tier-II reading. Scale-free event sizes do not force a $1/f$ temporal spectrum. The counterexample sharpens the relational claim: changing the readout and its duration weighting changes the exponent. The temporal null is not an embarrassment to be explained away; it is a mirror exposing which hypotheses the superposition theorem actually needs.

6. Contrasts and exclusions

6.1 Johnson–Nyquist noise

In the classical low-frequency lumped-resistor model, the one-sided voltage PSD is

$$S_V(f) = 4k_B T R.$$

That registry contains no nested transfer hierarchy relevant to this readout. This is a scoped modeling statement, not a claim that a physical resistor possesses no microscopic modes. Circuit response and quantum corrections matter outside the classical band.

6.2 Independent sampling

Independent identically distributed samples give a white discrete-time spectrum under the usual stationary assumptions. No cross-scale transfer relation is registered.

6.3 Brownian motion

Ideal Brownian position is nonstationary; formally or under finite-observation/generalized-PSD treatment it carries f^{-2} because it integrates white noise. Scale invariance alone therefore does not identify the superposition mechanism, a discharge boundary, or flat logarithmic relaxation weight.

6.4 BTW temporal activity

This is the strongest internal contrast: nested threshold dynamics and scale-free event sizes coexist with a temporal spectrum near f^{-2} . It defeats any claim that P1–P4 automatically produce $1/f$ in every readout.

7. What is claimed

Claim set

1. Under the exact hypotheses of Theorem 3.1, effective weight $g(\tau) = A\tau^{-\gamma}$, $0 < \gamma < 2$, gives interior spectrum $S(f) \sim f^{\gamma-2}$.
2. Equal effective weight per logarithmic relaxation-time interval gives $1/f$ within the finite band, with crossover scales controlled by the cutoff times.
3. When a readout inherits continuous homogeneity without adding a scale, it is power-law under the regularity conditions of Proposition 4.1. Scale-free underlying structure alone does not guarantee that inheritance.
4. Different readouts of one registered configuration may carry different exponents. The exponent belongs to the coupled configuration/readout relation.
5. The five domains display corresponding nested, bounded, scale-organized relations. This is a Tier-II structural correspondence, not a claim of common microscopic cause or mutual reduction.
6. Mathematical arrows used for calculation do not independently establish causal or ontological fundamentality.
7. A geometrically spaced finite registry ladder with equal effective weight per logarithmic cell converges, in its dense-ladder limit, to the finite-band $1/f$ form derived in §11; finite spacing may retain log-periodic ripple.

Tier III direction. Transfers need not begin when a readout distinguishes an event. Eventhood depends on the observational relation and resolution scale. This claim remains open to the approach-signature problem in §9.

Not claimed. The paper does not claim that $1/f$ uniquely diagnoses nested transfer; that every nested configuration gives $1/f$; that all five domains share one physical substrate; that the Cherenkov stable index is derived microscopically; or that any registry exhausts reality.

8. Exponent ledger

Case	Readout	Exponent	Status/source of exponent
Turbulence	$E(k)$	$-5/3$	K41 similarity plus dimensional analysis; not four-fifths law alone
Turbulence	$S_2(r)$	$+2/3$	K41 prediction; intermittency qualifies exact empirical scaling
Turbulence	$S(f)$, Taylor	$-5/3$	Linear coordinate map with constant PSD Jacobian
Relaxation superposition	$S(f)$	$\gamma - 2$	Theorem 3.1 in the interior finite band
Accretion disc	luminosity PSD	near -1	Lyubarskii diffusion model under stated radial-fluctuation assumptions
Trapped charge	current/voltage PSD	near -1	Theorem 3.1 when effective trap weight is flat in $\log \tau$

Case	Readout	Exponent	Status/source of exponent
Cherenkov LDF	fitted spatial tail	parameter-dependent	Empirical stable-density representation; radial convention must be retained
Seismicity	$N(> E)$	$-b/1.5$	Gutenberg–Richter plus energy–magnitude conversion
BTW sandpile	temporal PSD	near -2	Avalanche dynamics/lifetime scaling; not derived here from Theorem 3.1
Brownian position	generalized/finite PSD	-2	Integrated white noise; contrast case
Johnson noise	classical low-frequency PSD	0	Lumped equilibrium-noise model; contrast case

9. Observer, mirror, and self-definition

An observer-independent resolution criterion would need to specify when a transfer becomes an event without itself selecting a timescale, gate, or relation. This paper does not assume such an exterior criterion. Observer and observed are not interchangeable roles, but the observable distinction arises through their registered relation.

The mirror does not merely report a completed object. Let

$$\mathcal{W} = (O, M, F, \rho, N, \Lambda),$$

where O is the object as registered, M the contrasting description or mirror, F the frame, ρ the reflective relation, N the object’s enacted self-description, and Λ the residue. Neither object nor mirror supplies the independent ground of the other. Their contrast is evidence through difference, not an adversarial determination in which one description must eliminate the rest.

The same discipline applies to language. A mark becomes meaningful only by being distinguished and related under a gate. The closure that releases meaning also determines what the expression does not carry. In compact form,

$$\text{expression} = (\text{distinction}, \text{relation}, \text{frame}, \text{exclusion}, \text{residue}).$$

This does not weaken exact conclusions. It indexes them. A rigorous conclusion should terminate an ambiguity in the stated calculation without terminating inquiry into the reality for which the calculation serves as one description.

An approach signature for resolution itself remains open. Spectral crossovers provide a partial model: independently estimated cutoff times predict where the scaling band changes. A comparable precursor that announces the onset of distinguishability before a readout resolves it is not supplied here.

10. Falsification surfaces

The following surfaces separate theorem, empirical instantiation, and structural reading.

F1 — theorem/instantiation surface. A mathematical counterexample satisfying the hypotheses but not equation (2) refutes the theorem. An empirical PSD inconsistent with (2) within quantified

uncertainty falsifies the proposed instantiation: at least one claimed hypothesis, convention, or measurement relation does not hold as registered.

F2 — logarithmic control surface. Establish $g(\tau) \propto 1/\tau$, including readout coupling rather than number density alone, yet observe no finite-band behavior consistent with (4). This is the $\gamma = 1$ control case.

F3 — projection surface. Establish the exact homogeneity hypothesis of Proposition 4.1 over a continuous scaling range without an added scale, yet obtain a regular readout that is not homogeneous with the stated exponent.

F4 — domain-instantiation surface. In any worked case, show that the stated Tier-I formula, fitted parameter, or cited empirical relation is false under its declared convention. This defeats that registration and any Tier-II comparison depending on it; it does not automatically refute the independent superposition theorem.

F5 — structural-reading surface. Show that the proposed common organization can be assigned only by deleting a declared, outcome-relevant difference among the domains, or that the mapping cannot preserve nesting, bounded range, transfer measure, and readout dependence simultaneously. This defeats the five-domain unity claim without requiring a common microscopic test.

F6 — registry-discipline surface. Identify an argument whose conclusion requires promoting a registry-relative coordinate, boundary, or computational orientation into an unmarked hierarchy of fundamentality. Unless an additional witness for that promotion is supplied, the argument fails by the paper's own method.

F7 — registry-ladder surface. Construct the geometric ladder (5) with weights $A\Delta$ and Lorentzian kernels as stated, yet show that its Riemann sums do not converge to equation (4) as $\Delta \rightarrow 0$. This would refute Proposition 11.1. At fixed Δ , disagreement with an exact $1/f$ law is not by itself a refutation because discrete-scale ripple is permitted.

These surfaces do not claim that every conceivable criticism has been exhausted. They name finite locations at which the paper has put mathematical or structural weight.

11. Driven consequences: the registry ladder as a $1/f$ readout

Several questions posed by the preceding sections can be advanced without claiming a completed ontology. The key move is to treat the registry ladder itself as a finite-band family of registered relaxation scales and ask what spectrum its relational readout has.

11.1 A geometric ladder of frames

Let the characteristic times of successive registry stations be geometrically spaced:

$$\tau_j = \tau_{\min} e^{j\Delta}, \quad j = 0, 1, \dots, N, \quad N\Delta = \log \frac{\tau_{\max}}{\tau_{\min}}. \quad (5)$$

The exponential in (5) is not decorative. It converts additive steps in the index j into multiplicative steps in scale:

$$j \mapsto j + 1 \quad \Longleftrightarrow \quad \tau \mapsto e^\Delta \tau.$$

Thus Euler's e appears in two exact roles in the construction: a single registered level has exponential correlation $e^{-|t|/\tau}$, while equal increments of registry order trace a geometric ladder $\tau_j = \tau_{\min} e^{j\Delta}$. This does not make e an ontological cause; it records the coordinate equivalence between additive traversal and multiplicative scale.

Assign effective spectral weight $w_j = A\Delta$ to each logarithmic cell. Its discrete readout is

$$S_{N,\Delta}(f) = \sum_{j=0}^N A\Delta \frac{2\tau_j}{1 + (2\pi f\tau_j)^2}. \quad (6)$$

Proposition 11.1 (Dense-ladder limit; Tier I). *With fixed positive endpoints $\tau_{\min}, \tau_{\max}$, equation (6) converges as $\Delta \rightarrow 0$ to*

$$S(f) = \int_{\tau_{\min}}^{\tau_{\max}} \frac{2\tau}{1 + (2\pi f\tau)^2} A \frac{d\tau}{\tau}.$$

Consequently,

$$S(f) = \frac{A}{\pi|f|} [\arctan(2\pi|f|\tau_{\max}) - \arctan(2\pi|f|\tau_{\min})],$$

and $S(f) \sim A/(2|f|)$ in the interior band.

Proof. From $\log \tau_j = \log \tau_{\min} + j\Delta$, equation (6) is a Riemann sum in $u = \log \tau$:

$$S_{N,\Delta}(f) = \sum_j A K(f, e^{u_j}) \Delta \longrightarrow \int_{\log \tau_{\min}}^{\log \tau_{\max}} A K(f, e^u) du.$$

Since $du = d\tau/\tau$, the result is the $\gamma = 1$ case of Theorem 3.1. $\square$

For a ladder with fixed spacing Δ and equal weight w per station, the corresponding continuum coefficient is $A = w/\Delta$. Finite spacing may produce log-periodic ripple about the $1/f$ envelope; it does not justify replacing the discrete sum by an exact continuum law without an error estimate.

11.2 Why stopping produces both meaning and finite power

The stopping discipline is not merely rhetorical. For a stationary readout, total variance is the integral of its PSD. An ideal $1/f$ law across all positive frequencies would give

$$\int_0^\infty \frac{C}{f} df = \infty,$$

with logarithmic divergence at both endpoints. Likewise, infinitely many equal-variance registry stations carry divergent total variance even if the pointwise geometrically spaced spectral sum remains finite at a fixed nonzero f .

Finite scoping supplies $\tau_{\min}$ and $\tau_{\max}$. Those same limits make the aggregate finite and generate the low- and high-frequency crossovers. Thus

$$\text{local closure} \quad \nleftrightarrow \quad \text{finite observable band}.$$

Removing the cutoffs may express potential iteration, but it does not produce a single normalizable

stationary readout with equal weight at every scale. The potentially unbounded registry and each finite readable registration therefore remain neither identical nor independent.

11.3 Ordered comparison without an exterior simultaneous view

Let F_j denote the frame resolving station j , with local readout R_j . A comparison requires a declared transition or translation

$$T_{j \rightarrow k} : R_j \mapsto R_k.$$

The family $\{F_j, T_{j \rightarrow k}\}$ can be traversed and iterated without presuming an observer position that resolves every frame simultaneously. A later comparison frame may register F_j and F_k together, but it is itself another station with its own resolution and residue. Iteration can enlarge comparison; it cannot silently manufacture an unindexed outside.

This gives a precise version of the non-simultaneity claim: frames may be separately closed and ordered for comparison, while the comparison is an additional registered operation rather than an observer-free view of all closures at once.

11.4 Perturbation as the self/other seam

Let x_s and x_o denote channels registered in self and other roles, and suppose the observable is sensitive to their relative displacement. In the simplest linearized form,

$$y = L(x_s - x_o).$$

Its spectrum is

$$S_y(f) = |L(f)|^2 [S_{ss}(f) + S_{oo}(f) - 2 \operatorname{Re} S_{so}(f)]. \quad (7)$$

If the channels coincide perfectly under this gate, $y = 0$: the relation supplies no contrast for that readout. Perturbation makes the seam resolvable. What appears “unclean” relative to an identity gate is therefore the signal-bearing relative term, not automatically contamination external to the structure.

Equation (7) also prevents an overclaim. Correlations can cancel or reinforce scale contributions; they may change normalization, corners, or exponent. A $1/f$ relational readout follows when the *effective residual weight*, including the cross-spectral term, is approximately flat per logarithmic relaxation interval. The signature belongs to the coupled self/other/readout relation, not to either channel in isolation.

11.5 Structural consequence

The registry and the $1/f$ theorem now meet in a derived statement:

Registry-ladder consequence

A finitely scoped, geometrically spaced ladder of exponentially correlated registry stations converges to a $1/f$ interior readout in the dense-ladder limit when its effective relational weight is equal per logarithmic scale interval; a sufficiently fine discrete ladder approximates that envelope and may retain log-periodic ripple. Its finite stopping points become the spectral crossovers. Extending the ladder enlarges the band; it does not eliminate the need for a scoped readout. Perturbation supplies the relative contrast through which the ladder becomes observable, while correlations determine whether the equal-log weighting survives.

This result does not show that every recursive registry must produce $1/f$. It shows that the registry construction already contains an explicit sufficient route to the signature, and it identifies the exact places where that route can fail: non-geometric spacing, unequal residual weight, non-Lorentzian local kernels, scale-changing filters, or cross-spectral cancellation.

12. Residue ledger and executable continuations

This section is not a catalogue of questions awaiting a single terminal verdict. It separates what the present construction already decides from the relations that a subsequent traversal would have to carry.

12.1 Distributed calibration, not shared experience

An empirical comparison is a registry-to-registry construction. If observers A and B use calibration maps C_A, C_B and a declared transition map $T_{A \rightarrow B}$, the communicable path has the form

$$x_A \mapsto C_A(x_A) = q_A \mapsto T_{A \rightarrow B}(q_A) = q_B.$$

The shared object is a calibrated referent or checksum, not an assertion that the observers share an unmediated experience. For Theorem 3.1, the stronger empirical test therefore estimates the effective weight $g(\tau)$ along one calibrated registry path and the spectrum $S(f)$ along another, then compares them through a stated transition map. Inferring both from the same spectral fit would make premise and confirmation registry-identical.

12.2 Correlated contributions: an explicit exponent rule

For a difference readout $y = s - o$, equation (7) gives

$$S_y(f) = S_{ss}(f) + S_{oo}(f) - 2 \operatorname{Re} S_{so}(f).$$

Suppose, on a common band,

$$S_{ss} \sim A_s f^{-\alpha_s}, \quad S_{oo} \sim A_o f^{-\alpha_o}, \quad \operatorname{Re} S_{so} \sim A_c f^{-\alpha_c}.$$

The dominant uncanceled term fixes the observed exponent. If $\alpha_s = \alpha_o = \alpha_c = \alpha$, then

$$S_y(f) \sim (A_s + A_o - 2A_c) f^{-\alpha}.$$

Correlation changes the normalization while preserving the exponent unless the leading coefficient cancels; a frequency-dependent transfer filter can also change the exponent. Thus preservation and alteration are both admitted, but only after the comparison rule and cancellation condition are declared.

12.3 Three operational readout classes

The relevant distinction is not a universal division among objects. It is a classification of readouts under a stated scale action:

$$\begin{aligned} R_{\text{homoc}}(\lambda x) &= \lambda^{-\kappa} R_{\text{homoc}}(x) && \text{(continuous homogeneity),} \\ R_{\text{break}}(x) &= x^{-\kappa} \Phi(x/x_0) && \text{(an introduced scale or break),} \\ R_{\text{inv}}(x) &= x^{-\kappa} P\left(\frac{\log x}{\log q}\right), \quad P(z+1) = P(z) && \text{(discrete scale invariance).} \end{aligned}$$

These are observable and describable distinctions within the chosen registry. They do not promote one class into an ontologically prior object.

12.4 A one-step Cherenkov tail construction

The stable-law fit in §5.3 can be connected to a finite-moment local kernel by a scale mixture. Consider the one-dimensional Gaussian mixture

$$Q(r) = \int_0^\infty \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{r^2}{2\sigma^2}\right) p(\sigma) d\sigma. \quad (8)$$

If the distribution of cascade widths satisfies

$$p(\sigma) \sim C\sigma^{-(1+\alpha)} \quad (\sigma \rightarrow \infty),$$

then the substitution $u = r/\sigma$ gives

$$\begin{aligned} Q(r) &\sim \frac{C}{\sqrt{2\pi}} r^{-(1+\alpha)} \int_0^\infty u^\alpha e^{-u^2/2} du \\ &= \frac{C 2^{(\alpha-1)/2}}{\sqrt{2\pi}} \Gamma\left(\frac{1+\alpha}{2}\right) r^{-(1+\alpha)}. \end{aligned}$$

Hence $\alpha \simeq 1.3$ corresponds to both $p(\sigma) \sim \sigma^{-2.3}$ and a one-dimensional tail $Q(r) \sim r^{-2.3}$. The heavy tail may therefore arise from the distribution across cascade scales even though every conditional Gaussian kernel has finite moments.

This is a sufficient construction, not a first-principles derivation of the measured α . It relocates the missing calculation to a named object:

$$\text{cascade generations} \longrightarrow p(\sigma) \longrightarrow \alpha.$$

The convention matters: a radial density per unit area carries an additional geometric Jacobian, so its reported exponent must not be identified with the exponent of the one-dimensional density in equation (8) without conversion.

12.5 Resolution is indexed

A resolution threshold is to be stated with its detector, calibration, timescale, and decision gate. No unindexed or “observer-free” boundary is required by the construction. Changing any of those coordinates changes the resolving relation; it does not reveal a hidden boundary that existed independently in the same descriptive form.

12.6 Discrete and continuous scale registration

The operational comparison is between a geometric ladder $\tau_j = \tau_0 q^j$ and a continuous measure $g(\tau) d\tau$, not between a “genuine” and an illusory discreteness. Finite geometric spacing can produce log-periodic modulation of a power-law envelope. Within declared comparison classes, such modulation favors discrete-scale registration; its absence does not prove a continuum, because smoothing, finite resolution, or phase averaging can erase it.

12.7 A partial transition-map result

Let $\tau' = \phi(\tau)$ be differentiable and monotone. The logarithmic measure transforms as

$$d \log \tau' = \frac{\tau \phi'(\tau)}{\phi(\tau)} d \log \tau.$$

It is therefore preserved up to a constant exactly when $\tau \phi'(\tau)/\phi(\tau) = a$, whose positive solutions are

$$\phi(\tau) = C\tau^a.$$

Power maps preserve flatness in logarithmic scale up to the constant factor a ; a general transition map reweights the scale measure and can manufacture an apparent exponent. The executable continuation is to identify the admissible ϕ in each worked domain rather than posit one universal transition.

12.8 Recursive frame edges remain plural

No singular representation of every future frame edge is expected. At each extension, the registry can declare which prior residue has become internal, which degrees of freedom have been added, and which new edge the extension presents. Infinite potential and indefinite iteration are thereby registered together: neither is installed as the completed cause or fundamental ground of the other.

The paper closes one finite-band theorem, qualifies its projection principle, and presents five domains as mirrors of a nested-transfer organization. It does not close the realities described by those domains. Object, mirror, frame, transformation, and residue form the offered wholeness. Differences among exponents and mechanisms prevent that wholeness from collapsing into a single adversarial determination.

These continuations do not reopen the paper because it failed to close. They record what its closure has made calculable. The remaining residue is not a list of unanswered prompts demanding one verdict; it is the set of relations not carried by the present traversal. Exactness ends the present ambiguity. It does not end the subject.

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